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Counters and comparators (Exercise made by AI)

Learning Objectives:

- Understand the operation of **up-counting (CTU)** and **down-counting (CTD)** counters in TIA Portal.
- Learn to use **comparators** ($=$, $>$, $<$, etc.) to control logic based on counter values.
- Practice using both **global memory (MW)** and **direct data block (DB)** references in comparisons.
- Build a simple PLC application simulating a production line scenario.

✚ PART 1: Up-Counter (CTU) with Comparator (MW)

Scenario:

Products pass through a sensor on a conveyor. Each product increases the counter value. When the count exceeds 10, a signal light turns on.

Instructions:

1. Create a **CTU counter**.
2. Store the counter value in **MW10** using a MOVE instruction (e.g., MOVE CTU.ACC → MW10).
3. Use a comparator: $MW10 > 10 \rightarrow Q0.0 = \text{TRUE}$.
4. Add a **reset button** to clear the counter.

🔄 PART 2: Down-Counter (CTD) with Comparator (DB)

Scenario:

Products are removed from storage. Each removal decreases the counter value. When the remaining quantity drops below 3, a buzzer is activated.

Instructions:

1. Create a **CTD counter**.
2. Use the counter's **data block value** (e.g., DB1.ACC) in a comparator.
3. $DB1.ACC < 3 \rightarrow Q0.1 = \text{TRUE}$.
4. Add a **reset button** to restore the counter to its initial value.

🔄 PART 3: Combined Counter Logic

Scenario:

Simulate a warehouse system where products are added (CTU) and removed (CTD). Calculate the current stock level and trigger alerts based on thresholds.

Instructions:

1. Use both **CTU and CTD counters**.
2. Calculate stock level: $MW20 = CTU.ACC - CTD.ACC$.
3. Add comparators:
 - $MW20 > 20 \rightarrow Q0.2 = \text{"Overstock alert"}$
 - $MW20 < 5 \rightarrow Q0.3 = \text{"Low stock alert"}$
4. Add reset buttons for both counters.

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